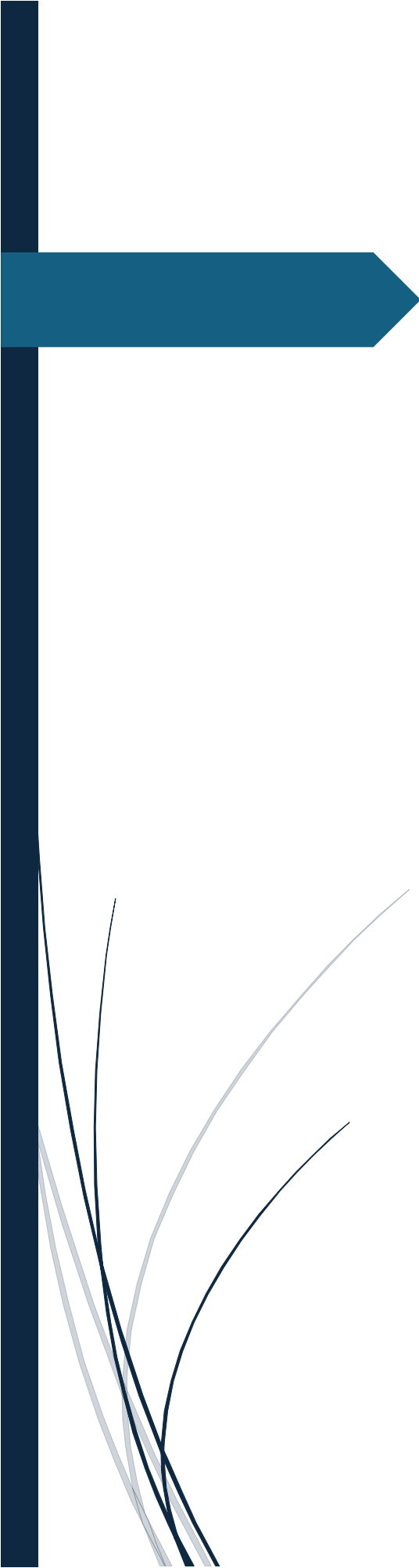




Assignment 3

Basic Network Hardware Theory

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.

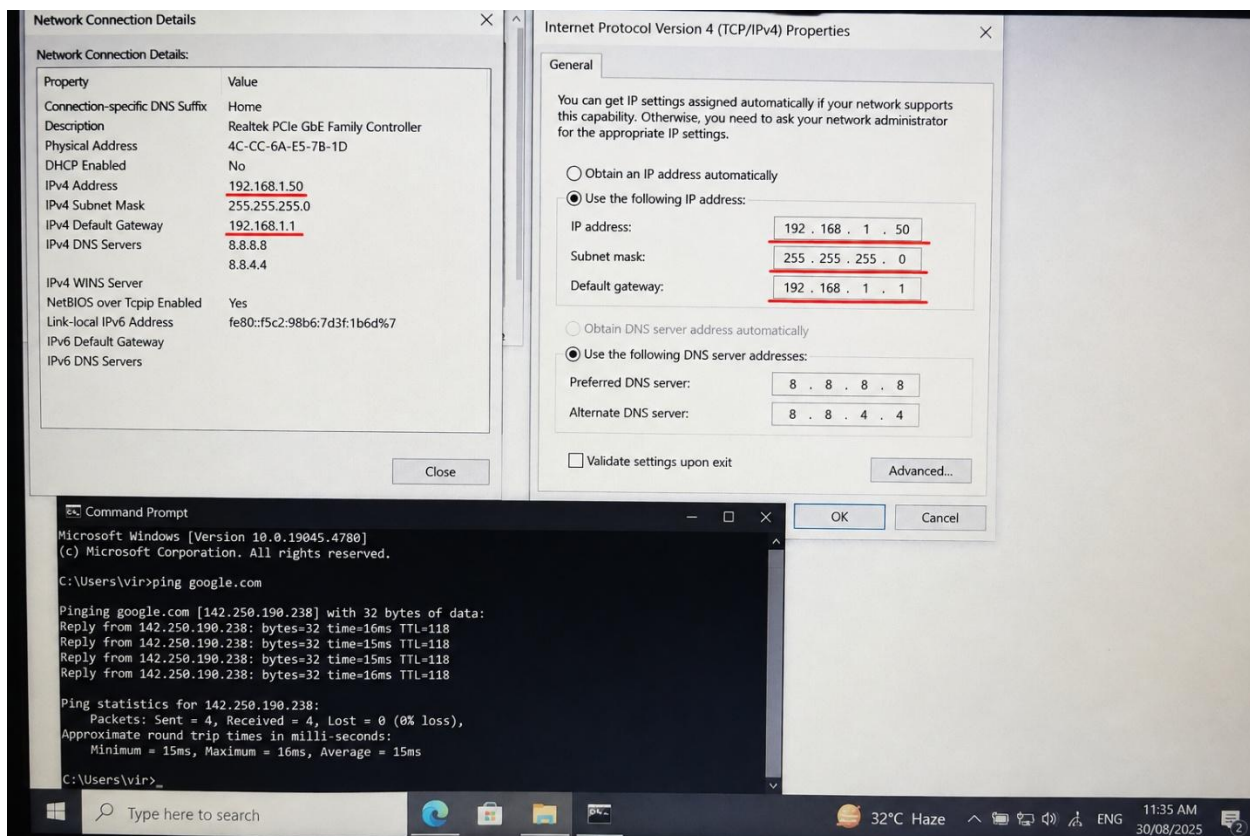
A login form for an Airtel router. At the top left is the "airtel" logo. At the top right is a language selector set to "English". Below the logo is a "Username" label followed by a text input field. Below that is a "Password" label followed by a password input field. At the bottom of the form is a grey "Sign in" button.

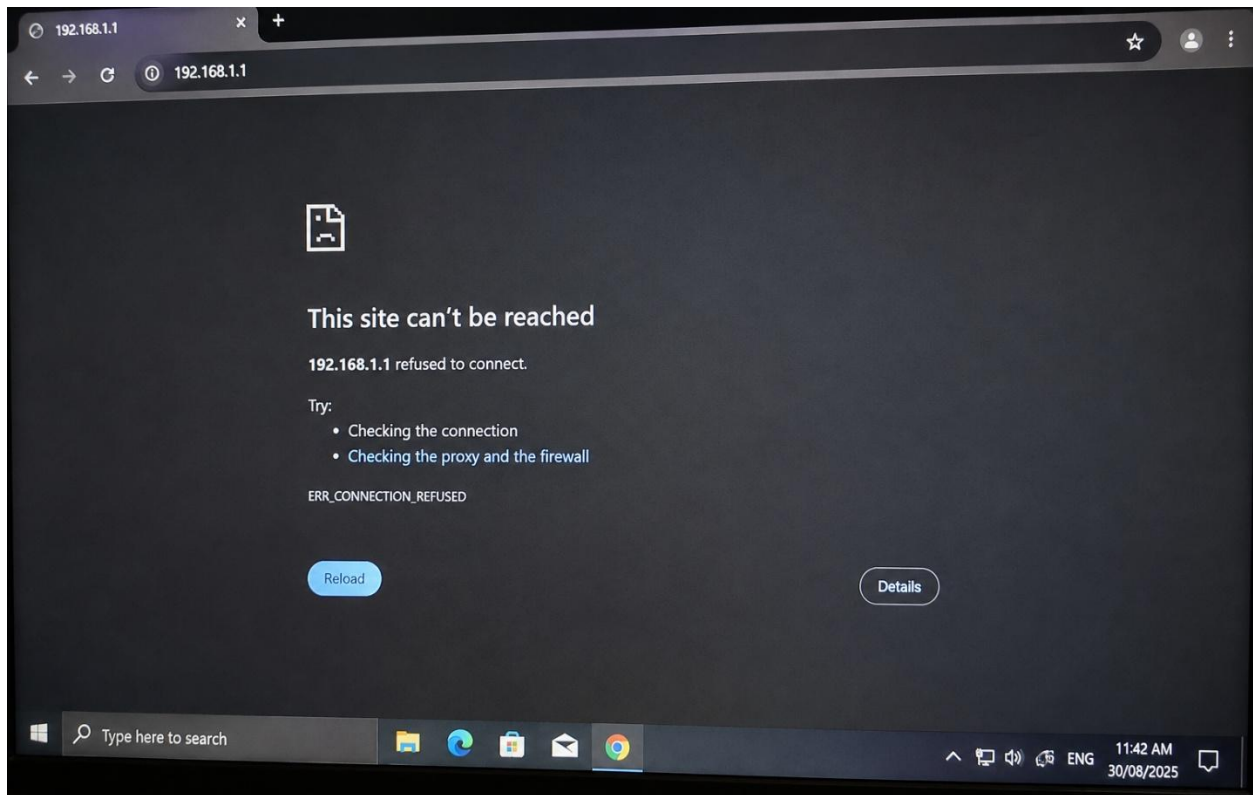
2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

Network Detail	Your Laptop
IP Address	192.168.1.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1

When my laptop is connected to the Wi-Fi network, it is assigned an **IP address of 192.168.1.2**. The **subnet mask 255.255.255.0** identifies the local network, while the **default gateway 192.168.1.1** is the router that connects my laptop to the internet.

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access. Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.





steps:

1. Open **Control Panel** → **Network Connections**.
2. Right-click your **Wi-Fi/Ethernet** → **Properties**.
3. Select **Internet Protocol Version 4 (TCP/IPv4)** → **Properties**.
4. Select **Use the following IP address**.
5. Enter:
 - IP: **192.168.1.50**
 - Subnet Mask: **255.255.255.0**
 - Gateway: **192.168.1.1**
6. Click **OK** and test your internet.
7. **After testing**, go back and select **Obtain an IP address automatically**.
8. Click **OK**.

4. In your router's **admin** panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

➤ **DHCP Range:** 192.168.1.2 – 192.168.1.254

➤ The **DHCP range** is the group of IP addresses that the router can automatically give to devices such as phones, laptops, and TVs.

➤ If **two devices get the same IP address**, an **IP conflict** occurs. This can cause the devices to lose internet access or have problems connecting to the router and other devices. The router normally prevents this by giving each device a different IP address.